

SHA-256² Dual-BARRIER Architecture

Barrier 2: Alignment Randomization

2^{260} entropy (information-theoretic)

Attack

Barrier 1: AES-256 Encryption

2^{256} security

**Genomic
Data**

Two mathematically independent
protection layers:
Both must be broken simultaneously

Combined Security: $2^{256} \times 2^{260} = 2^{516}$ bits

